

PROBLEM-BASED ASSIGNMENT

Development of a Heuristic Function for Seminar Hall Allocation

Artificial Intelligence

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1. Introduction

Artificial Intelligence can be used to solve problems where there may be many possible solutions. Seminar hall allocation is one such problem. A college may have several seminar halls with different seating capacities, facilities and locations. Guest lectures, workshops, project reviews, department meetings and student club programmes all need a suitable hall at a suitable time.

An allocation system should not only look for a hall that is free. A free hall may be too small, too large, far from the organizing department or may not have the facilities required for the event. Therefore, the system needs a way to estimate how suitable each hall is for a request. A heuristic function provides this estimate.

2. Problem Statement

The problem is to develop a heuristic function for selecting a suitable seminar hall for a seminar request inside a college. The system should evaluate each possible hall using factors such as unused capacity, missing facilities, booking clashes and the distance from the organizing department.

Instead of checking every possible combination of seminars and halls, the heuristic function assigns a score to each candidate allocation. The search process can then use the score to prefer allocations that appear more suitable. Since the objective is to minimize the cost of the allocation, a lower heuristic value represents a better allocation.

3. Need for the Proposed System

- Colleges may have several halls that can host the same event.
- The first available hall is not always the most suitable hall.

- A very large hall for a small event wastes space and cooling and lighting resources.
- A hall without the required equipment can disturb the event.
- Manual allocation can lead to double booking and repeated rescheduling.
- A heuristic provides a simple way to compare alternative halls.

4. Scope

The proposed approach can be used for allocating seminar halls, auditoriums and similar shared venues in a college. It can be extended later by adding information such as priority of events, maintenance schedules, accessibility requirements and past usage data when such information is available.

5. Objectives

The main objectives of the seminar hall allocation heuristic are:

- To find a suitable hall for a given seminar request.
- To reduce the number of empty seats in the allocated hall.
- To provide the facilities required by the event.
- To avoid booking clashes with other events.
- To reduce the travelling distance for the organizers and participants.
- To provide a good allocation without checking every possible allocation.

6. Input

The system can use the following information:

- Seminar request details, such as event name, date and time slot.
- Expected number of participants, such as 120.
- Facilities required, such as projector, microphone system and air conditioning.
- Capacity of each seminar hall.
- Facilities available in each hall.
- Existing bookings of each hall.
- Distance from the organizing department to each hall.

7. Representation of the Allocation Problem

The problem can be represented as a graph with two groups of nodes. One group contains the seminar requests and the other group contains the halls. A line between a request and a hall means

that the hall is being considered for that request. Each hall stores information such as capacity, facilities, existing bookings and distance.

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Seminar Request ---- Hall A
|
|----- Hall B
|
|----- Hall C
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An example of hall information is shown below for a request on a given day.

Hall	Capacity	Facilities	Booked Slots	Distance from Department
Hall A	150	Projector, Mic, AC	2:00 PM - 4:00 PM	200 m
Hall B	130	Projector, AC	None	100 m
Hall C	200	Projector, Mic, AC	10:00 AM - 12:00 PM	350 m

For example, a seminar for 120 participants may be allocated to Hall A, Hall B or Hall C. The heuristic function evaluates each candidate allocation rather than relying on only one measurement.

8. Assumptions

- The halls and their capacities, facilities and existing bookings are known to the system.
- Expected participant count and required facilities are given in the request.
- Halls that are too small for the event are not considered.
- Missing facilities and clashes can be represented using numerical penalty values.
- The allocation is considered a minimization problem.
- The weights used in the heuristic can be adjusted according to the college requirement.

9. Heuristic Function

A heuristic function assigns a numerical score to a candidate allocation. The score combines the factors that are important for seminar hall allocation. The function used in this assignment is:

$$H(A) = W1(U) + W2(M) + W3(CI) + W4(Dist)$$

Where:

- A = Candidate allocation (a seminar request assigned to a hall).
- U = Unused seats, calculated as hall capacity minus expected participants.
- M = Missing facilities, the number of required facilities not available in the hall.

- CI = Clash penalty, the number of existing bookings that overlap with the requested time slot.
- Dist = Distance from the organizing department to the hall, in metres.
- W1, W2, W3 and W4 = Weights assigned to the four factors.

For the example in this assignment, the weights are selected as follows:

$$H(A) = 2(U) + 20(M) + 100(CI) + 1(Dist)$$

When many seminars are allocated together, the heuristic value of the complete allocation is the sum of the values of all seminar-hall pairs.

Explanation of the Terms

Unused seats are included because a hall much larger than the audience wastes space and resources. Missing facilities are treated as a penalty because an event without the required equipment may not run properly. A clash is given the largest weight because two events cannot use the same hall at the same time, so a hall with a clash should almost never be selected. Distance is included because a nearer hall reduces the travelling effort of organizers and participants.

The weights determine the relative importance of the factors. In this example, clash and missing facilities receive larger penalty weights so that a hall is not selected only because it is close or has a suitable size.

10. Why a Heuristic is Useful

- It reduces the amount of search needed when many halls and requests are involved.
- It combines several practical factors into one score.
- It allows the system to compare halls using the same measurement.
- It can be used with search methods such as Hill Climbing.
- It can be modified when the requirements of the college change.

11. Interpretation

Because this is a minimization problem, a lower value of $H(A)$ indicates a more suitable allocation according to the selected criteria. The heuristic does not claim that the allocation is mathematically perfect; it gives a practical estimate that helps the search process make a better choice.

12. Worked Example

Suppose a department wants to organize a seminar for 120 participants. The event requires a projector, a microphone system and air conditioning. Two halls are free in the requested time slot, Hall A and Hall B. Their estimated values are shown below.

Hall	Unused Seats U	Missing Facilities M	Clash CI	Distance (m)	Heuristic
Hall A	30	0	0	200	260
Hall B	10	1	0	100	140

Hall A has a capacity of 150 and all three facilities. Hall B has a capacity of 130 and has a projector and air conditioning, but no microphone system.

Hall A Calculation

$$H(A) = 2(30) + 20(0) + 100(0) + 1(200)$$

$$H(A) = 60 + 0 + 0 + 200$$

$$H(A) = 260$$

Hall B Calculation

$$H(B) = 2(10) + 20(1) + 100(0) + 1(100)$$

$$H(B) = 20 + 20 + 0 + 100$$

$$H(B) = 140$$

The calculations show that Hall B has the lower heuristic value. Although Hall B has a missing facility, its smaller number of unused seats and shorter distance result in the lower total score.

13. Comparison of the Halls

- Hall A has all the required facilities and no missing-facility penalty.
- Hall A has more unused seats and is farther from the department.
- Hall B is a closer hall and fits the audience more closely.
- Hall B has a missing-facility penalty because it has no microphone system.
- After combining all four factors, Hall B receives the lower heuristic value.
- Therefore, the search method would prefer Hall B for this example.

The example demonstrates why a heuristic can be more useful than considering only one factor. It also shows the importance of the weights: if the microphone system were essential, the weight of M should be increased, or the facility should be treated as a compulsory condition. In an actual implementation, the values and weights should be chosen using appropriate college data.

14. Hill Climbing Approach

A simple Hill Climbing method can be used with the heuristic function. Hill Climbing starts with an initial allocation and repeatedly looks for a neighboring allocation that has a better heuristic value.

A neighboring allocation is created by moving one seminar to a different hall or by exchanging the halls of two seminars.

15. Algorithm

1. Start with an initial allocation of all seminar requests to halls.
2. Calculate the heuristic value of the current allocation.
3. Generate a neighboring allocation by changing the hall of one seminar.
4. Calculate the heuristic value of the neighboring allocation.
5. Compare the new value with the current value.
6. If the neighboring allocation has a lower value, accept it as the current allocation.
7. Continue until no better neighboring allocation is found.
8. Return the final allocation.

16. Pseudocode

START

Create an initial allocation of seminars to halls

Calculate $H(\text{current_allocation})$

REPEAT

 Generate a neighboring allocation

 Calculate $H(\text{neighbor_allocation})$

 IF $H(\text{neighbor_allocation}) < H(\text{current_allocation})$ THEN

 Accept neighbor_allocation as current_allocation

 ELSE

 Keep current_allocation

 END IF

UNTIL no better neighboring allocation exists

Display current_allocation

STOP

17. Working of the Method

- The system first selects an available allocation as the starting solution.
- It calculates the allocation's heuristic score.
- A nearby alternative allocation is generated by changing one hall.

- The alternative allocation is evaluated using the same heuristic function.
- If the new allocation has a lower score, the system moves to that allocation.
- The process continues until an improvement cannot be found.

18. Limitation of Hill Climbing

Hill Climbing can stop at a local optimum. This means that it may find an allocation that cannot be improved by moving a single seminar even though another arrangement of several seminars could have a lower overall score. Therefore, the quality of the final result depends on the starting allocation and the neighboring allocations considered.

19. Advantages

- The heuristic function is simple and easy to calculate.
- Multiple factors can be considered at the same time.
- The method can reduce unnecessary searching among hall combinations.
- The weights can be changed according to college requirements.
- The approach can be adapted to different numbers of halls and events.
- The method can support a basic allocation system without requiring every possible allocation to be examined.

20. Limitations

- The result depends on the quality of the selected weights.
- Expected participant counts may change after the request is made.
- A low score may still hide a missing facility that the event badly needs.
- Hill Climbing may reach a local optimum.
- The system needs updated information when bookings or halls are changed.

21. Possible Improvements

- Treat capacity and clash-free booking as compulsory conditions.
- Add event priority so that important events are considered first.
- Add maintenance and cleaning time between two bookings.
- Add accessibility information such as ramps and lifts.
- Use a more advanced search strategy when many halls and events are involved.

22. Conclusion

A heuristic function provides a practical way to solve the seminar hall allocation problem. A college can have many halls that could host the same event, and selecting a hall only because it is free may not give the most suitable result. By combining unused capacity, missing facilities, booking clashes and distance, the proposed heuristic provides a single score that can be used to compare allocations.

In this assignment, a lower heuristic value represents a more suitable allocation. The example demonstrates how two halls can be evaluated using the same formula. A Hill Climbing approach can then use this heuristic to repeatedly improve an allocation until no better neighboring allocation is found.

The proposed approach is simple enough for a basic college seminar hall allocation system and can be extended when additional information becomes available. Overall, heuristic-based allocation demonstrates how Artificial Intelligence can be applied to a practical problem in a college environment.

23. Summary

The assignment developed a heuristic function for seminar hall allocation, defined the required inputs and objectives, demonstrated the calculation using two halls, and described a Hill Climbing method and pseudocode for improving the selected allocation.